

Probability Basics Assignment

Question 1: A die is rolled. What is the probability of getting:

- (a) An even number
- (b) A number greater than 4

Answer- Total Possible Outcomes: When a standard six-sided die is rolled, the sample space S is :

$$S = \{1, 2, 3, 4, 5, 6\}$$

The total number of outcomes, $n(S) = 6$.

(a) Probability of getting an even number

Even numbers on a die are: $\{2, 4, 6\}$

Number of favourable outcomes = 3

Using the formula:

Probability = (Number of favourable outcomes) / (Total number of outcomes)

$$P(\text{Even number}) = 3 / 6$$

$$P(\text{Even number}) = 1 / 2$$

Answer: The probability of getting an even number is $1/2$.

(b) Probability of getting a number greater than 4

Numbers greater than 4 are: $\{5, 6\}$

Number of favourable outcomes = 2

$$P(\text{Number} > 4) = 2 / 6$$

$$P(\text{Number} > 4) = 1 / 3$$

The probability of getting a number greater than 4 is $1/3$.

Question 2: In a class of 50 students:

20 like Mathematics (M)

15 like Science (S)

5 like both subjects

What is the probability that a student chosen at random likes Mathematics or Science?

Answer- Number of students who like Mathematics (M) = 20

Number of students who like Science (S) = 15

Number of students who like both subjects ($M \cap S$) = 5

Total number of students = 50

Using the formula:

$$n(M \cup S) = n(M) + n(S) - n(M \cap S)$$

$$n(M \cup S) = 20 + 15 - 5$$

$$n(M \cup S) = 30$$

So, the number of students who like Mathematics or Science = 30

$$\text{So, } P(M \cup S) = 30 / 50$$

$$P(M \cup S) = 3 / 5$$

The probability that a student chosen at random likes Mathematics or Science is $3/5$.

Question 3: A bag has 3 red and 2 blue balls. If one ball is drawn randomly and is red, what is the probability that the next ball is also red (without replacement)?

Answer- A bag contains 3 red balls and 2 blue balls.

Total number of balls = 5

It is given that the first ball drawn is red and it is not replaced.

After drawing one red ball:

$$\text{Remaining red balls} = 3 - 1 = 2$$

$$\text{Remaining total balls} = 5 - 1 = 4$$

Now, Probability = (Number of favourable outcomes) / (Total number of remaining outcomes)

$$P(\text{Second ball is red} \mid \text{First ball is red}) = 2 / 4$$

$$P = 1 / 2$$

The probability that the next ball is also red (without replacement) is $1/2$.

Question 4: The population of a school is divided into 60% boys and 40% girls. If you want equal representation of both genders in the sample, which method should you use: Simple Random Sampling or Stratified Sampling? Why?

Answer- If we use Simple Random Sampling, students are selected completely at random. This means the sample will most likely reflect the population proportion (around 60% boys and 40% girls). It does not guarantee equal representation.

Hence,

In Stratified Sampling, the population is divided into groups (called strata) based on a characteristic — here, gender (boys and girls). Then, we can select an equal number of students from each group.

Since the requirement is equal representation of both genders, Stratified Sampling is the correct method because it allows us to control how many students are chosen from each group.

Question 5: The average height of 1000 students = 160 cm. A sample of 100 students shows an average height = 158 cm. Find the sampling error.

Answer- Population mean (μ) = 160 cm
Sample mean ($\bar{x}$) = 158 cm

Formula:

$$\text{Sampling Error} = \text{Sample Mean} - \text{Population Mean}$$

Substituting the values:

$$\text{Sampling Error} = 158 - 160$$

$$\text{Sampling Error} = -2 \text{ cm}$$

Interpretation:

The negative sign shows that the sample mean is 2 cm less than the population mean.

Question 6: The population mean salary is ₹50,000 with $\sigma = ₹5,000$. If we take a sample of 100 employees, what is the standard error of the mean (SEM)?

Answer- The Population mean salary (μ) = ₹50,000
Standard deviation (σ) = ₹5,000
Sample size (n) = 100

Formula would be :

$$\text{Standard Error of the Mean (SEM)} = \sigma / \sqrt{n}$$

$$SEM = 5000 / \sqrt{100}$$

$$\sqrt{100} = 10$$

$$SEM = 5000 / 10$$

$$SEM = 500$$

The Standard Error of the Mean (SEM) = ₹500.

Question 7: In a group of 100 students: 40 like Cricket (C) 30 like Football (F) 10 like both Cricket and Football Find the probability that a student likes at least one sport.

Answer- In a group of 100 students:

Students who like Cricket (C) = 40

Students who like Football (F) = 30

Students who like both Cricket and Football ($C \cap F$) = 10

Step 1: Use the formula

Number of students who like at least one sport:

$$n(C \cup F) = n(C) + n(F) - n(C \cap F)$$

Step 2: Substitute the values

$$n(C \cup F) = 40 + 30 - 10$$

$$n(C \cup F) = 60$$

So, 60 students like at least one sport.

Step 3: Calculate probability

Probability = (Number of favourable outcomes) / (Total number of students)

$$P(\text{At least one sport}) = 60 / 100$$

$$P = 0.6$$

The probability that a student likes at least one sport is 0.6 or $3/5$.

Question 8: From a deck of 52 cards, two cards are drawn without replacement. What is the probability that both are Aces?

Answer- Given in question

Total cards in a deck = 52

Total Aces = 4

Cards drawn = 2 (without replacement)

Step 1: Probability that the first card is an Ace

$$P(A_1) = 4/52$$

Step 2: Probability that the second card is an Ace (after one Ace is already drawn)

Remaining Aces = 3

Remaining cards = 51

$$P(A_2|A_1) = 3/51$$

Step 3: Multiply probabilities

$$\begin{aligned} P(\text{both cards are Aces}) &= (4 / 52) \times (3/51) \\ &= 12 / 2652 \\ &= 1 / 221 \end{aligned}$$

So, the $P(\text{Both cards are Aces}) = 1 / 221$

Question 9: A factory produces bulbs with 2% defective rate. If 5 bulbs are chosen at random, what is the probability that all are non-defective?

Answer- A factory produces bulbs with a 2% defective rate.

Defective probability = 2% = 0.02

non-defective probability = $1 - 0.02 = 0.98$

Number of bulbs chosen = 5

Step 1: Probability that one bulb is non-defective

$$P(\text{Non-defective}) = 0.98$$

Step 2: Probability that all 5 bulbs are non-defective

Since the selections are independent:

$$P(\text{all 5 non-defective}) = (0.98)^5$$

Step 3: Calculate

$$0.98^5 = 0.9039 \text{ (approximately)}$$

$$P(\text{all bulbs are non-defective}) = 0.9039$$

Therefore, the probability that all 5 bulbs are non-defective is approximately 0.9039 (or 90.39%).

Question 10: Differentiate between discrete and continuous random variables with examples

Answer-

Basis	Discrete Random Variable	Continuous Random Variable
Definition	A random variable that can take countable and distinct values .	A random variable that can take any value within a given range .
Nature of Values	Values are usually whole numbers and can be counted.	Values can be fractions or decimals and are measured.
Number of Possible Values	Finite or countably infinite number of values.	Infinite number of possible values within an interval.
Example	Number of students in a class, number of cars in a parking lot, number of heads in coin tosses.	Height of a person, weight of a student, temperature of a room.

Conclusion

A discrete random variable represents countable values, while a continuous random variable represents measurable values that can take any value within a range.